

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	41.8737	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	78.95798	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1382.6094	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	273.0781	1	0	0	0.962039	15.5
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	187.2745	1	0	0	0.965079	12.12
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.070523	0.070523	232.10132	2	0	0	0.628661	14.855

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 41.8737.
- Accepted epoch count: 0.

- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 78.95798.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1382.6094.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.

- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 273.0781.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 187.2745.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.066777.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070523.
- Held-out runtime ms: 232.10132.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.005029.
- Worst held-out instances:
 - X-n176-k26: feasible=True, penalized gap=0.098678, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.095455, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057661, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.05742, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.043401, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary

Condition	Feasibility (Heldout)	Feasible Gap	Penalized Gap	Runtime (ms)	Robustness Dispersion	Notes
Baseline Nearest Neighbor Constructive	1.0	0.2734	0.2734	41.87	0.0028	Fastest runtime, moderate gap
Baseline Clarke-Wright Savings	1.0	0.0738	0.0738	78.96	0.0078	Good gap, moderate runtime
Baseline Regret Insertion Local Search	1.0	0.4664	0.4664	1382.61	0.0869	Worst gap, slowest runtime
Direct Generate +1 Repair	0.0	0.0	3.0	273.08	0.0	No feasibility, high penalty
Budget-Matched No Replay	0.0	0.0	3.0	187.27	0.0	No feasibility, high penalty
Replay Solver Evolution (Best Heldout)	1.0	0.0705	0.0705	232.10	0.0050	Best feasible gap, good robustness, feasible

Conservative Interpretation

- **Direct synthesis (Direct Generate +1 Repair)** showed no feasibility on heldout, with high penalized gap.

- **Budget-matched independent search without replay** also failed to produce feasible solutions.
- **Replay-aware iterative search (Replay Solver Evolution)** achieved full feasibility with the best heldout feasible gap (0.0705), outperforming all fixed baselines on objective gap except Clarke-Wright (which was close at 0.0738) while maintaining good robustness and moderate runtime.
- Fixed baselines all had 100% feasibility; Clarke-Wright Savings had the best gap among them before replay.
- Replay clearly **beat** the budget-matched no-replay control, delivering feasibility and much lower gaps.
- Runtime for replay was higher than Clarke-Wright but substantially lower than Regret Insertion Local Search.
- Robustness (dispersion) was low for replay and baselines, indicating stable performance across held-out structure families.

Summary

- **Replay Solver Evolution** is the only learned condition that beats or matches fixed baselines in feasibility, objective gap, and robustness.
- **Replay-aware iterative search** clearly surpasses **budget-matched independent search** without replay in both feasibility and solution quality.